The model framework

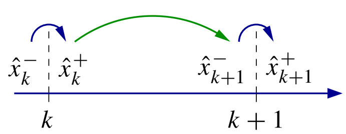
- This week, we derive the Kalman filter algorithm, which is a special case of sequential probabilistic inference.
- We will start by assuming that our system has a discrete-time general (possibly) nonlinear state-space model:

$$
\begin{aligned}
x_k &= f(x_{k-1}, u_{k-1}, w_{k-1}) \\
z_k &= h(x_k, u_k, v_k),
\end{aligned}
$$

  where we desire to estimate the present state x_k based on measurements made until this time $\mathbb{Z}_k = \{z_0, \ldots, z_k\}$.
- The development in this lesson is general; we will specialize to a linear model in later lessons this week and will reuse the general development in Course 3, Nonlinear Kalman Filters.

Vector notation

- We begin the derivation by defining some math notation that we will use from now on:
  - Superscript "$-$" indicates a predicted quantity based only on past measurements.
  - Superscript "$+$" indicates an estimated quantity based on both past and present measurements.
  - Symbol "$\hat{}$" indicates a predicted or estimated quantity: $\hat{x}^+$ or $\hat{x}^-$.
  - Symbol "$\tilde{}$" indicates an error: the difference between a true and predicted or estimated quantity: $\tilde{x} = x - \hat{x}$.
- So, one sampling-time interval looks like:



Matrix notation

- Symbol "Σ" is used to denote correlation between the two arguments in its subscript (autocorrelation if only one is given):

$$\Sigma_{xz} = \mathbb{E}[xz^T] \quad \text{and} \quad \Sigma_x = \mathbb{E}[xx^T] \quad .$$

- Furthermore, if the arguments are zero mean (as they often are in the quantities we talk about), then this represents covariance:

$$\Sigma_{\tilde{x}\tilde{z}} = \mathbb{E}[\tilde{x}\tilde{z}^T] = \mathbb{E}[(x - \mathbb{E}[x])(z - \mathbb{E}[z])^T],$$

  for zero-mean $\tilde{x}$ and $\tilde{z}$.

Cost function to minimize (optimize)

- Desire state estimate that minimizes "mean-squared error",

$$\begin{aligned}\hat{x}_k^{\mathrm{MMSE}}(\mathbb{Z}_k) &= \arg\min_{\hat{x}_k} \Big(\mathbb{E}\big[\, \|x_k - \hat{x}_k^+\|_2^2 \mid \mathbb{Z}_k \big] \Big) \\ &= \arg\min_{\hat{x}_k} \Big(\mathbb{E}\big[(x_k - \hat{x}_k^+)^T (x_k - \hat{x}_k^+) \mid \mathbb{Z}_k \big] \Big) \\ &= \arg\min_{\hat{x}_k} \Big(\mathbb{E}\big[x_k^T x_k - 2x_k^T \hat{x}_k^+ + (\hat{x}_k^+)^T \hat{x}_k^+ \mid \mathbb{Z}_k \big] \Big).\end{aligned}$$

- Solve for $\hat{x}_k^+$ by differentiating cost function and setting result to zero:

$$0 = \frac{\mathrm{d}}{\mathrm{d}\hat{x}_k^+} \mathbb{E}\big[x_k^T x_k - 2x_k^T \hat{x}_k^+ + (\hat{x}_k^+)^T \hat{x}_k^+ \mid \mathbb{Z}_k \big].$$

Preliminary solution to state estimate

- To differentiate the cost function, it is helpful to know the following identities from vector calculus,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}X} Y^T X = Y, \qquad \frac{\mathrm{d}}{\mathrm{d}X} X^T Y = Y, \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}X} X^T A X = (A + A^T) X.}$$

- Using these, we can solve for $\hat{x}_k^+$ in:

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}\hat{x}_k^+} \mathbb{E}\big[x_k^T x_k - 2x_k^T \hat{x}_k^+ + (\hat{x}_k^+)^T \hat{x}_k^+ \mid \mathbb{Z}_k \big] \\ 0 &= \mathbb{E}\big[-2(x_k - \hat{x}_k^+) \mid \mathbb{Z}_k \big] = 2\hat{x}_k^+ - 2\mathbb{E}\big[x_k \mid \mathbb{Z}_k \big] \\ \hat{x}_k^+ &= \mathbb{E}\big[x_k \mid \mathbb{Z}_k \big]. \end{aligned}$$

- So, we seek to develop an algorithm for computing or approximating $\mathbb{E}[x_k \mid \mathbb{Z}_k]$.

Prediction error and innovation

- Define <u>prediction error</u> $\tilde{x}_k^- = x_k - \hat{x}_k^-$ where we have also defined $\hat{x}_k^- = \mathbb{E}\big[x_k \mid \mathbb{Z}_{k-1} \big]$
  - Error is always "truth minus prediction" or "truth minus estimate."
  - We can't compute error in practice, since truth value is not known.
  - But, we can prove statistical results using this definition that give an algorithm for estimating the truth using measurable values.
- Also, define the measurement <u>innovation</u> (what is new or unexpected in the measurement) as $\tilde{z}_k = z_k - \hat{z}_k$, where $\hat{z}_k = \mathbb{E}\big[z_k \mid \mathbb{Z}_{k-1} \big]$.

Prediction error and innovation are zero mean

- Both $\tilde{x}_k^-$ and $\tilde{z}_k$ can be shown to be zero mean using method of iterated expectation: $\mathbb{E}_Z\big[\mathbb{E}_{X|Z}[X \mid Z]\big] = \mathbb{E}_X[X]$:

$$\mathbb{E}[\tilde{x}_k^-] = \mathbb{E}[x_k] - \mathbb{E}\big[\mathbb{E}[x_k \mid \mathbb{Z}_{k-1}]\big] = \mathbb{E}[x_k] - \mathbb{E}[x_k] = 0$$
$$\mathbb{E}[\tilde{z}_k] = \mathbb{E}[z_k] - \mathbb{E}\big[\mathbb{E}[z_k \mid \mathbb{Z}_{k-1}]\big] = \mathbb{E}[z_k] - \mathbb{E}[z_k] = 0.$$

- Note also that $\tilde{x}_k^-$ is uncorrelated with past measurements as they have already been incorporated into $\hat{x}_k^-$

$$\mathbb{E}[\tilde{x}_k^- \mid \mathbb{Z}_{k-1}] = \mathbb{E}\big[x_k - \mathbb{E}[x_k \mid \mathbb{Z}_{k-1}] \mid \mathbb{Z}_{k-1}\big] = 0 = \mathbb{E}[\tilde{x}_k^-].$$

Predict/correct solution

- Consider now expanding the relationship $\mathbb{E}[\tilde{x}_k^- \mid \mathbb{Z}_k]$:

$$\mathbb{E}[\tilde{x}_k^- \mid \mathbb{Z}_k] = \underbrace{\mathbb{E}[x_k \mid \mathbb{Z}_k]}_{\hat{x}_k^+} - \underbrace{\mathbb{E}[\hat{x}_k^- \mid \mathbb{Z}_k]}_{\hat{x}_k^-}.$$

  - This is true because $\hat{x}_k^- = \mathbb{E}[x_k \mid \mathbb{Z}_{k-1}]$ is a constant vector (it is not an RV), and further conditioning on $\mathbb{Z}_k$ has no additional effect.
- We can also expand this relationship a different way:

$$\mathbb{E}[\tilde{x}_k^- \mid \mathbb{Z}_k] = \mathbb{E}[\tilde{x}_k^- \mid \mathbb{Z}_{k-1}, z_k] = \mathbb{E}[\tilde{x}_k^- \mid z_k].$$

- Combining both expansions, we have $\hat{x}_k^+ = \hat{x}_k^- + \mathbb{E}[\tilde{x}_k^- \mid z_k]$, which is a predict/correct sequence of steps, as promised.

Summary

- You learned meaning of notation we will use, including superscripts "$-$" and "$+$", symbols "$\hat{}$" and "$\tilde{}$", and symbol "Σ".
- We are solving a minimum mean-squared-error problem to find a state estimate:
  - The initial solution was $\hat{x}_k^+ = \mathbb{E}[x_k \mid \mathbb{Z}_k]$.
  - We later refined this to be $\hat{x}_k^+ = \hat{x}_k^- + \mathbb{E}[\tilde{x}_k^- \mid z_k]$.
  - A predict/correct structure!
- But, what is $\mathbb{E}[\tilde{x}_k^- \mid z_k]$? That's what we look at next.